HW #7 — Moments and Review of Continuous RVs (25 pts)

MATH 241 — Probability, Fall 2026

Due Date: Monday, 11/9 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as one .pdf file.
- Responses can be hand-written or typed.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- Refer to syllabus for late work deductions.

Background

We will continue our focus on continuous random variables, using refined methods for finding items such as PDFs, CDFs, expected values, and variances. We will also look at finding quantiles and moments for continuous RVs.

Problems

(I) (10 pts) For each of the following random variables, I want you to find (a) $F^{-1}(y)$, (b) the first quartile, median, and third quartile, and (c) the third raw moment. You can start from the already-defined PDF/PMF and CDF formulas.

(1) $X \sim Exp(\lambda = 2)$

(2) $X \sim Unif(a = 0, b = 3)$

(3) $X \sim Geom(p = 0.2)$

- Do not worry about calculating the third raw moment for (3).

(II) (15 pts) Choose **five** problems on the study guide and work through these.

(III) (10 pts) Extra credit: Explore the following article from the Pew Research Center:

- Link: *Many Americans are struggling to afford basic necessities*

In this article, affordability among Americans is explored leading up to the 2026 midterms. Using this article, I want you to **summarize some results using random variables, probability distributions, and measures of center/spread**.

- Consider looking at some variable or a set of disjoint events.
- You might explore point/cumulative probabilities, moments, variance, etc. Provide work related to finding these items and show the results.
- Interpret the results in the context of affordability and the contents of the article.

Do not take more than two pages for this question. The work and analysis itself should be brief and concise.